

Hyperlink Guide - Milestone 5: Integration & Optimization

Hello reviewer! We recognize that our documentation is substantial and complex. In order to help guide you, we provide here a series of links and explanations relating to the milestone outputs, acceptance criteria and evidence of milestone completion for Milestone 5.

Please feel free to reach out if you have any questions.

Milestone Outputs

Full System Integration:

- Seamless integration of core components, including staking mechanisms, trustless bridge (with partners), and interoperability between Cardano and Bitcoin networks through Sundial's initial testnet.
 - This monorepo showcases all of Sundial's core components, including our btc lockers for staking, our trustless bridge integration with charms, and interoperability via our L2 system.
<https://github.com/sundial-protocol/sundial-monorepo>
- Unified infrastructure ensuring compatibility with UTxO-based assets and external DeFi protocols.
 - This monorepo also shows all infrastructure and is fully built and compatible with UTxO-based assets and Cardano, and therefore Cardano-based DeFi protocols
<https://github.com/sundial-protocol/sundial-monorepo>

Smart Contract Optimization:

- Performance enhancements to improve transaction efficiency and reduce gas costs.
 - The report attached highlights the improved fee and throughput optimisation test results:
<https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/12a24f9d9068d84a5925e482f42a04a5de84e0f9/report.md>

- This highlights the improvements made to the L2 smart contracts in terms of script size, execution units, and clarity:
<https://github.com/sundial-protocol/internal-docs/blob/main/reports/M5.5-Optimizations.md>
- Refinement of security measures to prevent exploits and inefficiencies.
 - [Hacken](#)
 - [Dual Defense](#)
 - [JSHy Review](#)
 - [SC Optimizations](#)

Enhanced User Experience & Interface:

- Fully functional and refined UI for staking, bridging, and DeFi interactions.
 - Fixed UX/UI feedback from users and added "remarks from developers" under with more technical details. Above the screenshot is feedback from users
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/125>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/126>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/127>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/128>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/129>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/130>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/131>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/132>
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 - <https://github.com/sundial-protocol/sundial-monorepo/issues/134>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/136>
- Improved onboarding process for new users with an intuitive design.
 - Onboarding guides have been published to a new page in the website:
<https://sundialprotocol.com/l2>

Acceptance criteria

Seamless Integration Across Sundial & Bitcoin Bridges:

- All components work as expected within bridging testnet environments.

- This monorepo showcases all of Sundial's core components, including our btc lockers for staking, our trustless bridge integration with charms, and interoperability via our L2 system.

<https://github.com/sundial-protocol/sundial-monorepo>

- Bridge transactions execute, maintaining speed and security.
 - This report gives an overview of the transactions involved in bridging with the integrated Charms bridge, and how to independently verify them.

<https://github.com/sundial-protocol/internal-docs/blob/main/reports/M5.2-Transactions-Report.md>

Optimized Smart Contracts for Cost & Performance Efficiency:

- Transaction costs are significantly reduced compared to Cardano mainnet examples.
 - The report attached highlights the improved fee and throughput optimisation test results:
- <https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/12a24f9d9068d84a5925e482f42a04a5de84e0f9/report.md>
- Performance benchmarks confirm faster processing speeds and seamless interactions.
 - The following report showcases a decreased number of blocks utilised to carry the same number of transaction on Cardano's L1, therefore showcasing higher transaction speeds, all with no issues occurring
- <https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/12a24f9d9068d84a5925e482f42a04a5de84e0f9/report.md>

Finalized UI with Positive User Feedback:

- User interface meets functional requirements and has been refined based on testing feedback.
 - Fixed UX/UI feedback from users and added "remarks from developers" under with more technical details. above the screenshot is feedback from users
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/125>

- <https://github.com/sundial-protocol/sundial-monorepo/issues/126>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/127>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/128>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/129>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/130>
 - <https://github.com/sundial-protocol/sundial-monorepo/issues/131>
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 - <https://github.com/sundial-protocol/sundial-monorepo/issues/136>
- Early users report a smooth experience with minimal friction in onboarding and staking processes
 - The following feedback showcases user feedback from a pre-release testing phase with minimal issues in staking and onboarding:
 - <https://github.com/sundial-protocol/internal-docs/blob/main/reports/M5.7-User-Feedback-Report.md>
 - Raw Feedback: <https://docs.google.com/spreadsheets/d/1jDpcMSO6qKgXR9xIAi5pVp90wn-V-Lwh/edit?gid=1713320098#gid=1713320098>

Evidence of milestone completion

Integration Testing & System Performance Reports:

- Performance benchmarks made public alongside GitHub commits and changelogs detailing completed integrations, bug fixes, optimizations and protocol performance/functionality benchmarks
 - L2 Infrastructure <https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/12a24f9d9068d84a5925e482f42a04a5de84e0f9/report.md>
 - L2 Smart Contracts <https://github.com/sundial-protocol/internal-docs/blob/main/reports/M5.5-Optimizations.md>
- Testnet transaction reports demonstrating interoperability with Bitcoin and Cardano.

- This report gives an overview of the transactions involved in bridging with the integrated Charms bridge, and how to independently verify them.
<https://github.com/sundial-protocol/internal-docs/blob/main/reports/M5.7-Transactions-Report.md>
- Public demo videos showcasing end-to-end functionality of staking, bridging, and DeFi features.
 - Video walkthrough of our initial staking defi product:
<https://drive.google.com/file/d/1CYUnfys0KMo2fQ7ZiMtDrywPvobfWF3H/view?usp=sharing>
 - Video walkthrough showing the bridging flow and full integration:
https://drive.google.com/file/d/1UZUYcRQ2LZQYUwyLDfEnpXPOdj9toYTC/view?usp=drive_link

Smart Contract Optimization Results:

- Comparative benchmarking reports illustrating improvements in transaction costs and processing speeds.
 - The report attached highlights the improved fee and throughput optimisation test results:
<https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/12a24f9d9068d84a5925e482f42a04a5de84e0f9/report.md>
- Third-party reviews or audits confirming smart contract efficiency and security conducted by at minimum 2 external auditing entities.
 - [Hacken](#)
 - [Dual Defense](#)
 - [JSHy Review](#)
 - [SC Optimizations](#)
- Codebase updates with explanations of optimizations made for cost reduction.
 - The report attached highlights the actions taken to achieve improved fee and throughput optimisation:
<https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/12a24f9d9068d84a5925e482f42a04a5de84e0f9/report.md>

User Interface Validation & Feedback:

- Recorded user feedback sessions highlighting UI/UX improvements.
 - The following feedback showcases user feedback from a pre-release testing phase with minimal issues in staking and onboarding:
 - <https://github.com/sundial-protocol/internal-docs/blob/main/reports/M5.7-User-Feedback-Report.md>
 - Raw Feedback:
<https://docs.google.com/spreadsheets/d/1jDpcMSO6qKgXR9xIAi5pVp90wn-V-Lwh/edit?gid=1713320098#gid=1713320098>
- Screenshots, walkthrough videos, or live streams demonstrating the final UI.
 - Video walkthrough of our initial staking defi product:
<https://drive.google.com/file/d/1CYUnfys0KMo2fQ7ZiMtDrywPvobfWF3H/view?usp=sharing>
 - Video walkthrough showing the bridging flow and full integration:
https://drive.google.com/file/d/1UZUYcRQ2LZQYUwyLDfEnpXPOdj9toYT/view?usp=drive_link
- Community beta testing results showing a high usability rating and successful onboarding experiences.
 - The following feedback showcases user feedback from a pre-release testing phase with minimal issues in staking and onboarding:
 - <https://github.com/sundial-protocol/internal-docs/blob/main/reports/M5.7-User-Feedback-Report.md>
 - Raw Feedback:
<https://docs.google.com/spreadsheets/d/1jDpcMSO6qKgXR9xIAi5pVp90wn-V-Lwh/edit?gid=1713320098#gid=1713320098>

Reviewer Responses

Milestone Reviewer 3

Thanks for the detailed submission. We traced the evidence through to source and most of it checks out — the repos, commits, closed issues, and the Hacken audit all verify accurately.

1. Can you confirm the beacon-minting fix is live on the deployed testnet, not just present in source?

We can see that `validators/free_mint.ak` was changed on 2026-05-25 (commit 2bfa2e7) so that minting now requires a signature from `permissioned_vk`, replacing the previous unconditional-pass policy. We also confirmed that fix is included in the commit your monorepo's `charms-bridge` submodule is pinned to.

What we can't confirm from source alone is what's actually running. Could you share either the deployed script hash / policy ID for the beacon-minting validator, or point us to a specific on-chain testnet transaction where a beacon was minted - so we can verify the gated policy is in effect on the live deployment?

Good catch, the built scripts got excluded from our earlier commit. We've republished the built scripts ([a86de93](#)). The built script hash you're looking for is

`f5d9ef3aea339b55ddb3cdcfdcf0578ca586c1abfc7a334ddf395758`

2. Are any beacon NFTs minted under the previous policy still active?

Because beacon UTXOs are referenced rather than spent, they persist indefinitely once created. Changing the minting policy going forward wouldn't invalidate a beacon that was already minted while the old policy was in place. Can you confirm whether any pre-2026-05-25 beacons are still in circulation and referenceable by the main validator, and if so, what happens to them?

There are not currently any old beacons on the L2 testnet ledger. In cases where a minted beacon is determined to be faulty or malformed, the beacon can be burned by submitting a ``mint`` tx with value -1.

3. Could you update the PoA wording on the Charms-Cardano review?

The submission states the review's findings are "both resolved," but the linked document (`yHSJ/charms-independent-review`) still shows the high-severity finding as "Open" - it looks like the reviewer simply never updated the write-up after you shipped the fix. We're satisfied the fix itself is real; it's the citation that's inconsistent. Could you either have the reviewer update the document, or reword the entry to reference the fix commit directly and note that the reviewer's write-up predates it? We'd rather the evidence trail be self-consistent for anyone reading it later.

The reviewer has confirmed to us directly that both remediations are satisfactory. What hasn't happened is the write-up being updated - we've asked multiple times and it hasn't landed, and we don't think the milestone should hang on a third party's editing queue. So we'd like to take your second option and make the evidence stand on the commits:

The [independent code review](#) of [sundial-protocol/charms-cardano](#) was dated **2026-05-24** against commit [82264ef](#). It includes the following findings:

- one high-severity finding **§8.5**, main accepts forged beacons under the `free_mint` minting policy)
- one registration issue (**§9.2**, missing publish handler on `staking_groth16`).

Both were remediated after the review was written:

- **§9.2** by [c52584f](#) (**2026-05-23**)
- **§8.5** by [2bfa2e7](#) (**2026-05-25**)
- with follow-up tests in [770970a](#) (**2026-05-29**)

All commits are merged to [main](#).

4. What controls the `permissioned_vk` key?

The fix moves beacon minting from “permissionless” to “authorized by one key.” That’s a sound interim step, but it concentrates trust. Is that key held by a single party, a multisig, or something else — and is there a plan to move toward the governance-controlled or registry-based approach that was also discussed?

On testnet, this is a single keypair held by the dev team. On mainnet launch this will be a multisig of trusted charms developers who ratify a new version and put it on chain. Eventually we will move to decentralize this process further once the developer community is sufficiently large.

5. Which two entities satisfy the “minimum 2 external auditing entities” criterion, and is there any audit coverage of the Cardano-side validators?

Three separate questions here.

1. The submission links back to audit reports from the previous milestone. Since this milestone introduces new performance optimizations, we understand you would need to provide updated reviews/reports covering the optimizations.

The Locker and Charms reviews covered their performance optimizations, so they cover the optimized code.

The L2 contracts are built on Anastasia Labs' Midgard, and our changes are the optimizations documented in the M5.5 report. The Midgard contracts aren't mainnet-ready yet, and we'll commission an external audit once they're complete.

On timing: we're funding audits from a tight budget in a hard market, so auditing once against finished, optimized code rather than twice was the deliberate call.

2. On the entity count. Of the three engagements cited:

- Hacken — a formal audit. We verified its findings, severities, and resolution status against the published report; it all checks out.

- HackenProof (Dual Defense) — an ongoing bug-bounty program, and as we understand it Hacken's own platform rather than an independent organization.

- Josh Marchand — an independent review of charms-cardano, described in the document itself as a code review rather than a security audit.

The criterion permits "reviews or audits," so we're not disputing that a review can count toward it. Our question is simply which two entities you're counting, given Hacken and HackenProof appear to be the same organization.

We believe the HackenProof entity and audit process was sufficiently distinct to be considered a third entity/audit:

- Entirely different team of reviewers
- Entirely different process, with different portal and points of contact
- Included public reviews from over 100 hackers
- Entirely different internal development cycle, resulting in distinct versioning
- Included a finding resulting from a fix made in the Hacken audit

Regardless of whether you consider Dual Defense and Hacken one entity or two, the milestone requires 'at minimum 2 entities' & we believe the submitted reports are satisfactory.

3. We did not find any formal audit of Cardano smart contracts. Last but not least have you considered audit organisations such as: "Tx Pipe / Anastasia Labs" for those contracts? If such an audit has been provided (on Cardano contracts side), did we miss it? While you have committed to a review or an audit, we believe that a project of this magnitude requires an audit. While we are not going to refuse the milestone for this reason, we highly encourage you to increase from a review to an audit.

Yes, we will of course have a full audit of any mainnet code carrying significant funds. We have not completed all audits, but we feel we have completed enough to satisfy the requirements of

this milestone.

We have been in close contact with the teams you mentioned - in fact, most of the Cardano-facing smart contracts are developed with/by those teams. Anastasia Labs is one of the named partners on this project precisely because the L2 is built on their Midgard framework. Our optimisations are upstreamed to the AL team through appropriate channels.

6. Minor: could you point us at one or two concrete testnet transaction hashes?

The M5.2 Transactions Report describes the bridging flow well, but we'd like to verify a couple of the underlying transactions on a block explorer directly, since testnet transaction evidence is an explicit acceptance criterion. Any representative examples would do.

The deployed L2 is live at <https://rpc.testnet.sundialprotocol.com> (Cardano Preprod), health/ready → 200 {"status":"ready"}.

Block header 833445d629338f73a15f8f3ecf9f5a5eae30c6936e8eecfad6a00fc8 — contains 3 real transactions. All 3 are retrievable now with full CBOR.

The transactions

#	L2 tx hash	in → out	fee	notes
1	9ad083dcc3751ef2a50b3030261c6fcb499bd2a00f55154dde28971a0f045c0	genesis funder fa0ce700...#0 → 100 ADA + change	0.178333 ADA	300 B CBOR, 1 vkey witness, network_id: 0, valid
2	5dba244aef0a3327b463c541bb1ea0d199bfa4437a22ce29db47fde42ee10391	9ad083...#1 → 100 ADA + change	0.176925 ADA	272 B CBOR
3	eb40f88e892c1ca9aa0b4ae6dc8c73bd2825ca0bc271d6921e03c5c0f45106c3	5dba24...#1 → 100 ADA + change	0.176925 ADA	272 B CBOR